Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-I Business Mathematics Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part I Business Mathematics. It contains comments on candidates' responses to the 2019 HSSC - I Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E- Marking Notes

This includes overall comments on candidates' performance on every question and some specific examples of candidates 'responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

Key Observations

This year candidates faced difficulty in the questions based on Function, Equation of Straight Line and Discount and Sales. Candidates performed well on questions related to Matrices and Number systems.

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1:

a. Ms Zubaida paid Rs 4,750 for a suit after a discount of 15% followed by a sales tax of 10% of the actual price. Find the actual price of the suit.

b. It is given that $a:b=3:1$.

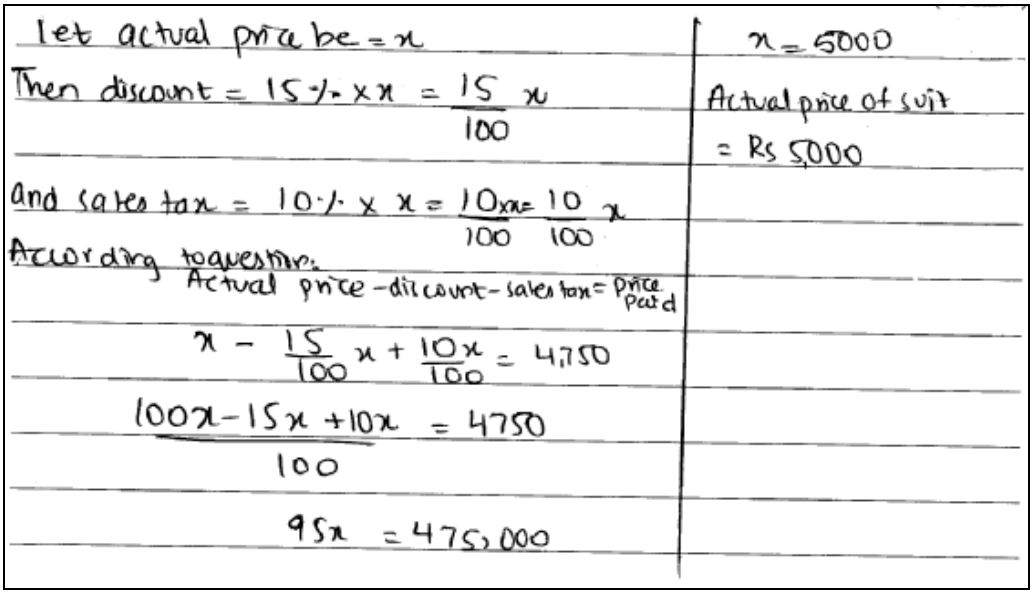
i. Express a in terms of b.

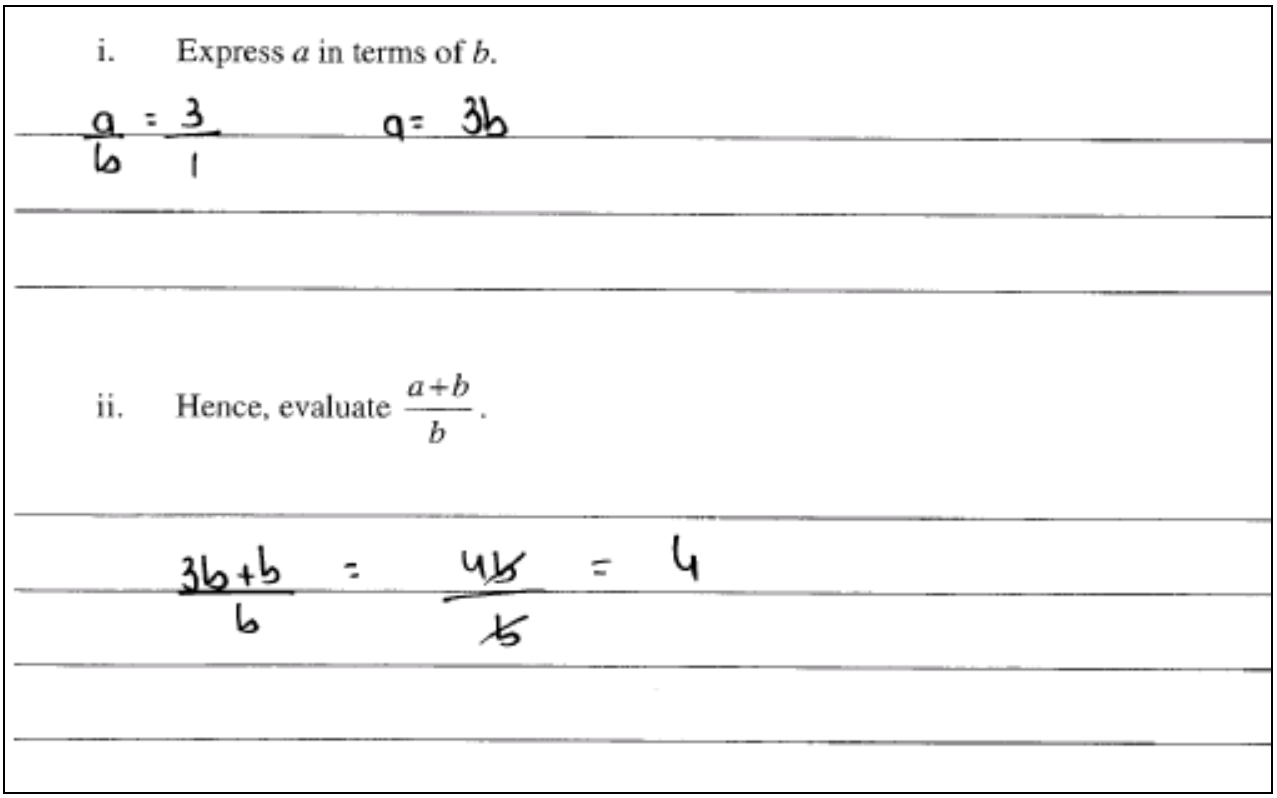
ii. Hence, evaluate $\frac{a+b}{b}$.

Better responses reported that candidates in part **a** correctly applied the given conditions and hence, they were able to manage the actual requirement of the question quite comfortably. This showed that the candidates had a good knowledge related to discount and sales tax.

Similarly, in part **b**, candidates successfully expressed a in terms of b and evaluated by writing $\frac{a}{b}+1$ and substituted the values accordingly.

Example:





Weaker responses revealed that candidates in part **a** lacked understanding about discount and sales tax. They did not use the conditions correctly and therefore, they were unable to connect the concept in finding the actual price of the suit. It was also seen that fewer candidates wrote 15% of 4,750 followed by 10% that was actually incorrect. Hence, they were stuck in proceeding further to get the required answer.

Similarly, in part **b**, candidates failed to express *a* in terms of *b* and hence, they were unable to evaluate the required form.

Example:

SP = 4750

Discount % = 15 %.

Sales tax = 10 %.

Actual price = ?

Solution:

$$MP = \frac{100 \times SP}{\text{remaining \%.}}$$

$$= \frac{100 \times 4750}{85}$$

$$MP = \frac{475000}{85} = 5588$$

Sales tax = 10% of MP

$$= \frac{10}{100} \times 5588$$

$$= 558.8 = 559$$

Thus MP = 5588 + 559

= 6147 Rs

$$\text{Actual price} = \frac{100 \times MP}{\text{Discount \%.}}$$

$$= \frac{100 \times 6147}{15}$$

= 40980 Rs.

i. Express a in terms of b.

$$\frac{a}{b} = \frac{3}{1}$$

or $a : b :: 3 : 1$ Ans

ii. Hence, evaluate $\frac{a+b}{b}$.

$$\frac{a+b}{b} = \frac{3+1}{1}$$

$$\text{Ans} = \frac{a+b}{b} = \frac{4}{1} \text{ Ans}$$

Question 2:

This question offered a choice between part **a** and part **b**. Majority of the candidates attempted part **a**.

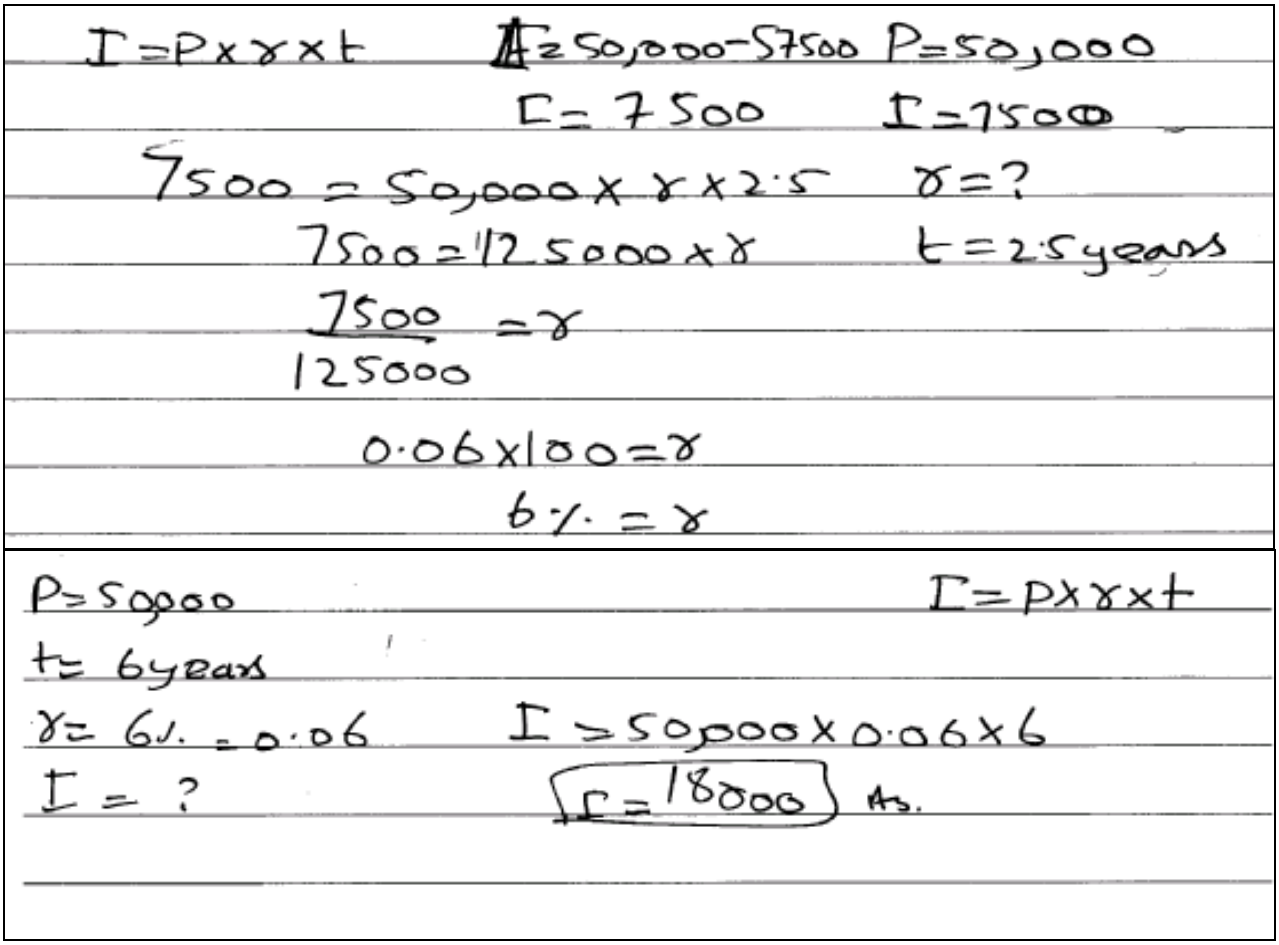
Question 2a:

i. Amina deposited Rs 50,000 in a saving scheme on annual simple interest basis. After 30 months, the amount deposited becomes Rs 57,500. Find the rate of simple interest.

ii. What will be the amount of interest Amina gets after 6 years at the same rate as in part (i)?

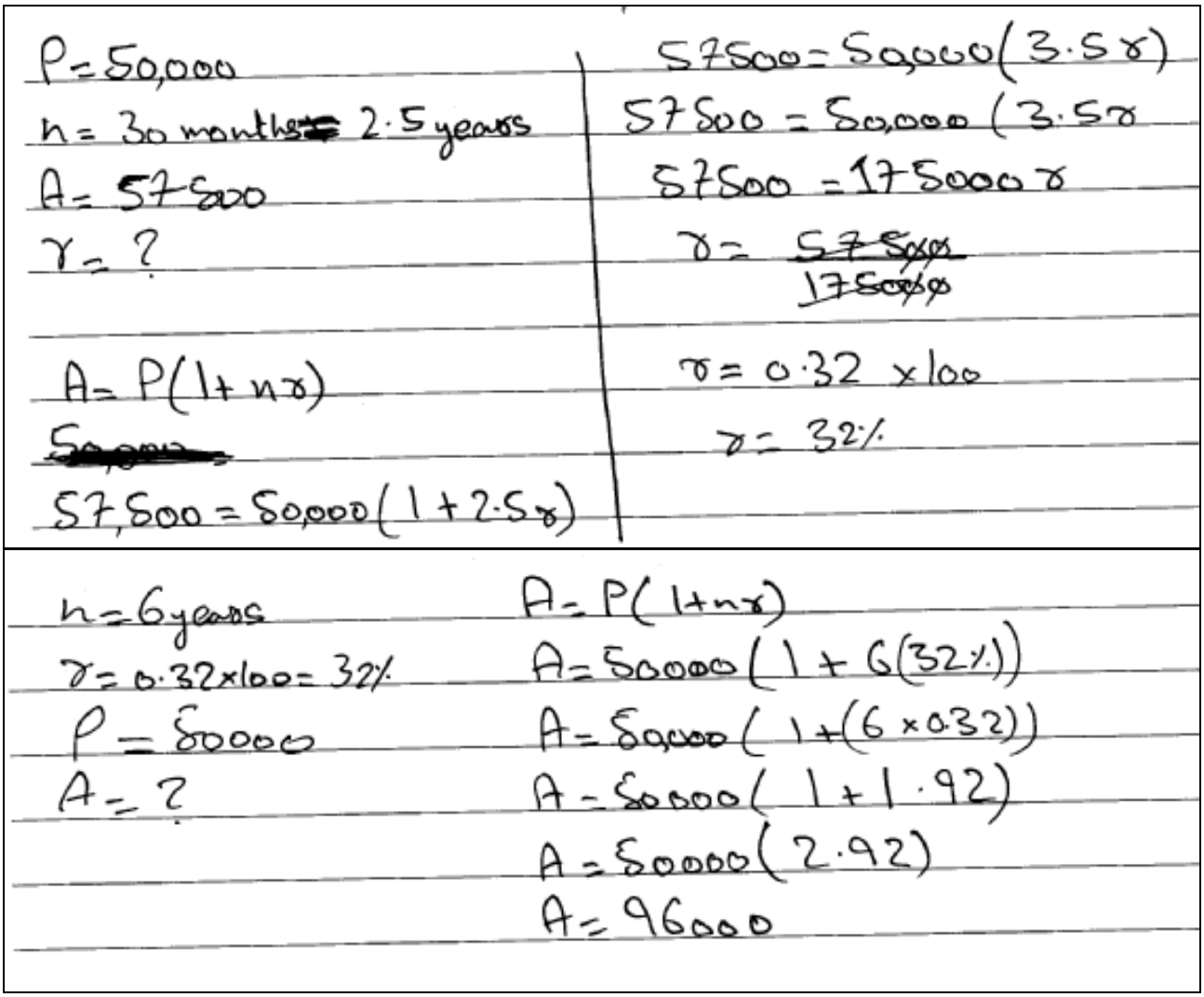
Better responses informed that the candidates in both the parts applied their knowledge to calculate the interest rate based on the concept of simple interest. It was noticed that candidates were able to comprehend the question well and used correct formula to get the required answer which highlighted the candidates' ability of attempting the question.

Example:



Weaker responses indicated that candidates were unable to differentiate between formula of simple interest and compound interest. Some weaker responses reported that candidates failed to find the amount of interest rate because they either made calculation errors or unable to use the rate of simple interest from part **i.**

Example:



Question 2b:

For a certain amount *x*, the compound interest compounded annually is Rs 100 more than the simple interest. The rate of simple and compound interest for 4 years is 6% per annum.

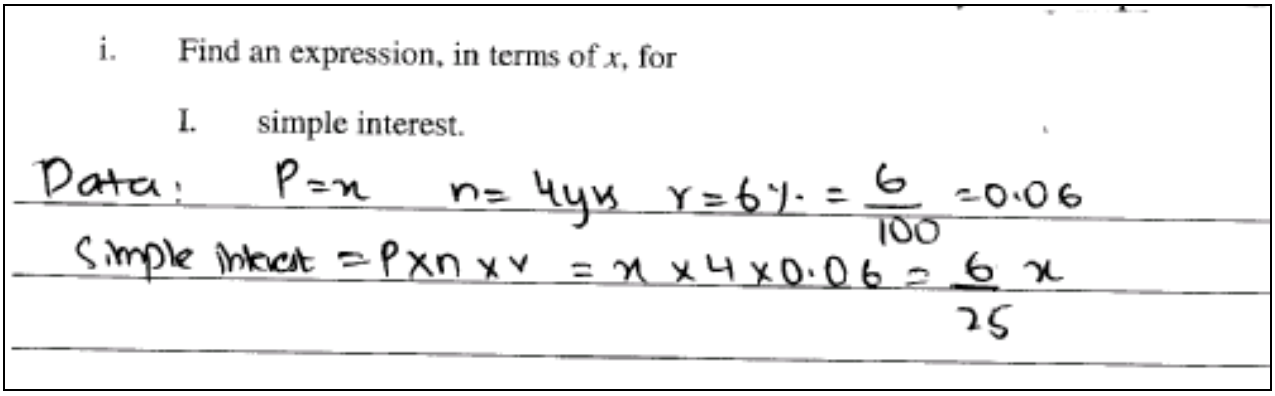
i. Find an expression, in terms of *x*, for

   I. simple interest.
   II. compound interest.

ii. Hence, find *x*.

Better responses revealed that candidates determined the expressions for simple and compound interest in terms of *x*. This helped them in using the relation correctly given in the question based on simple and compound interest to get the value of *x* required in the question.

Example:


i. Find an expression, in terms of x, for

I. simple interest.

Data: $P = x$ $n = 4$ yrs $r = 6\% = \frac{6}{100} = 0.06$

Simple interest $= P \times n \times r = x \times 4 \times 0.06 = \frac{6}{25}x$

II. compound interest. (2 Marks)

Data: $P = x$ $m = 1$ $i = 6\% = 0.06$ $t = 4yr$ $I = ?$

Solution: $A = P\left(1 + \frac{i}{m}\right)^{mt} = x\left(1 + \frac{0.06}{1}\right)^{1\times4} = x\,(1.06)^4$

$A = 1.262x$

$I = A - P = 1.262x - x = 0.262x$

compound interest = ~~Compound interest~~ $0.262x$

ii. Hence, find x. (2 Marks)

Since compound interest = 100 + simple interest

$0.262x = 100 + \frac{6}{25}x$

$0.262x - \frac{6}{25}x = 100$

$\frac{6.55 - 6}{25}x = 100$

$0.55x = 2500$

$\boxed{x = \frac{50,000}{11}}$ Ans

Weaker responses indicated that candidates were unable to grasp the concept based on simple and compound interest. They were confused in making the expressions for simple and compound interest. It was noticed that even though they found the expressions for simple and compound interest but made mistakes in translating the given condition into mathematical statement. They wrote $I = 100 + C$ instead of $C = 100 + I$ which was actually incorrect. As a result, they could not get the right answer.

Example:

i. Find an expression, in terms of x, for

I. simple interést.

$I = P.n.r$

$100 \times 4 \times 0.06$

24

II. compound interest.

$A = P(1+r)^n$

$100(1+0.06)^4$

$126.247696.$

ii. Hence, find x.

$A = P(1+nr)$

$A = 100(1+4 \times 0.06)$

$x = 124.$

Question 3:

This question offered a choice between part **a** and part **b**. Both the parts were attempted equally.

Question 3a:

It is given that *x* and *y* are binary numbers, where $x = 1011$ and $y = 1101$.

i. Without using a calculator, show that the value of $(x + y)$ in base 10 is 24.
ii. Hence, find $(x + y)_5 - 14_5$

Better responses reported that the candidates in part i comfortably converted the given binary number in base 10 after correct calculation of $(x + y)$. Similarly, in part ii, they easily converted into base 5. This helped them to reach the desired answer after the calculation of $(x + y)_5 - 14_5$. These responses by the candidates demonstrated a clear understanding of the concept being examined.

Example:

i. Without using a calculator, show that the value of $(x+y)$ in base 10 is 24.

$x=(1011)_2 = 1\times2^3+0\times2^2+1\times2^1+1\times2^0$

$= 8+0+2+1$

$= x=(11)_{10}$

$y=(1101)_2 = 1\times2^3+1\times2^2+0\times2^1+1\times2^0$

$= 8+4+0+1$

$y=(13)_{10}$

$(x+y) = 11 + 13 = 24$ — Proved

$(24)_{10} = (24)_{10}$

ii. Hence, find $(x+y)_5 - 14_5$

(x)

5	11
2	−1

$(21)_5$

(y)

5	13
2	−3

$(23)_5$

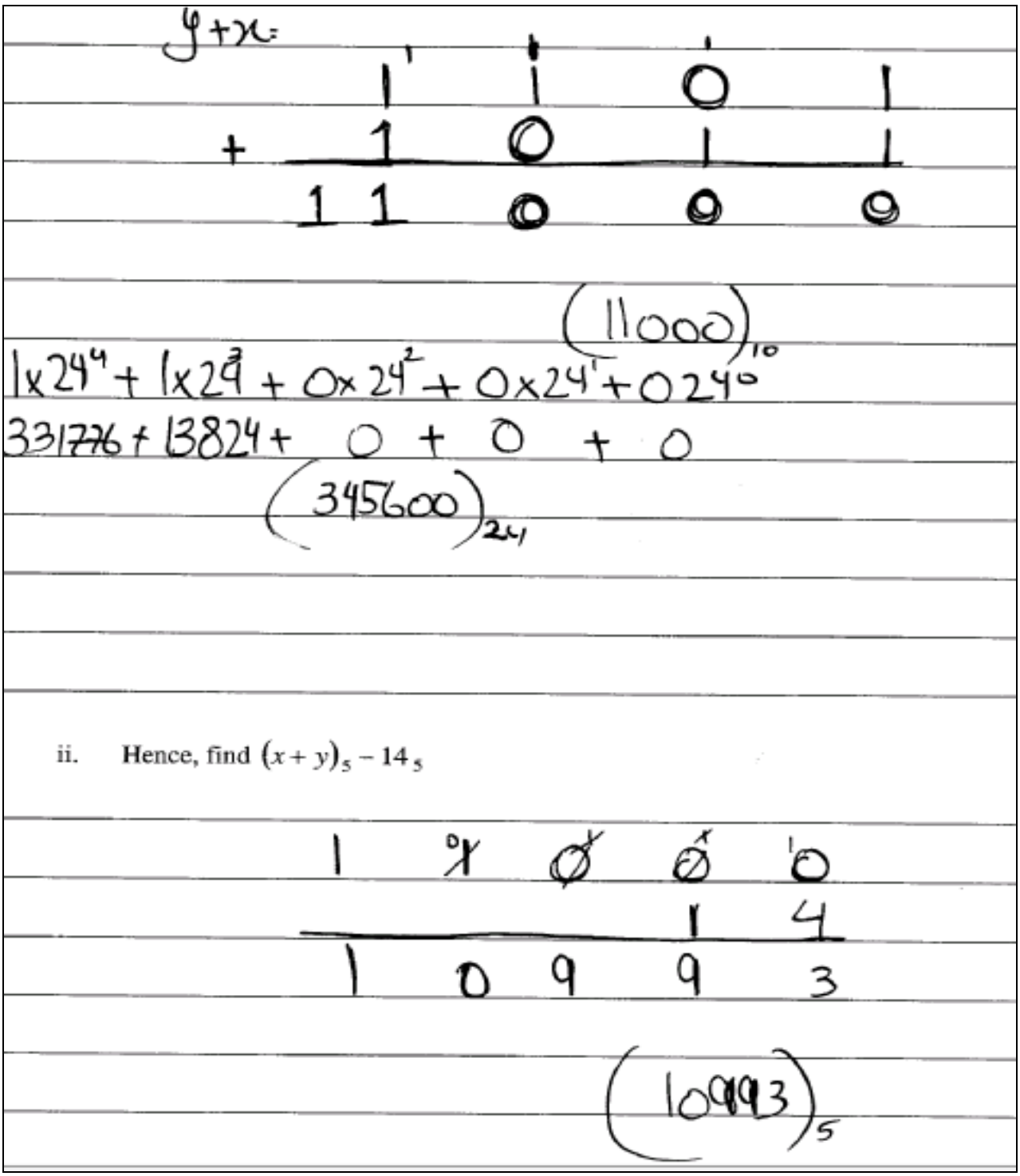
$(x+y)_5 = 21 + 23 = 44$

$(x+y)_5 - 14_5 = 44 - 14 = 30$

(30_5) Ans.

Weaker responses reflected that candidates faced difficulty in finding the conversion of number system. It was seen that they were not comfortable in conversion of base 10 to base 5 number system and even failed to proceed further in subtraction of base 5 number system.

Example:

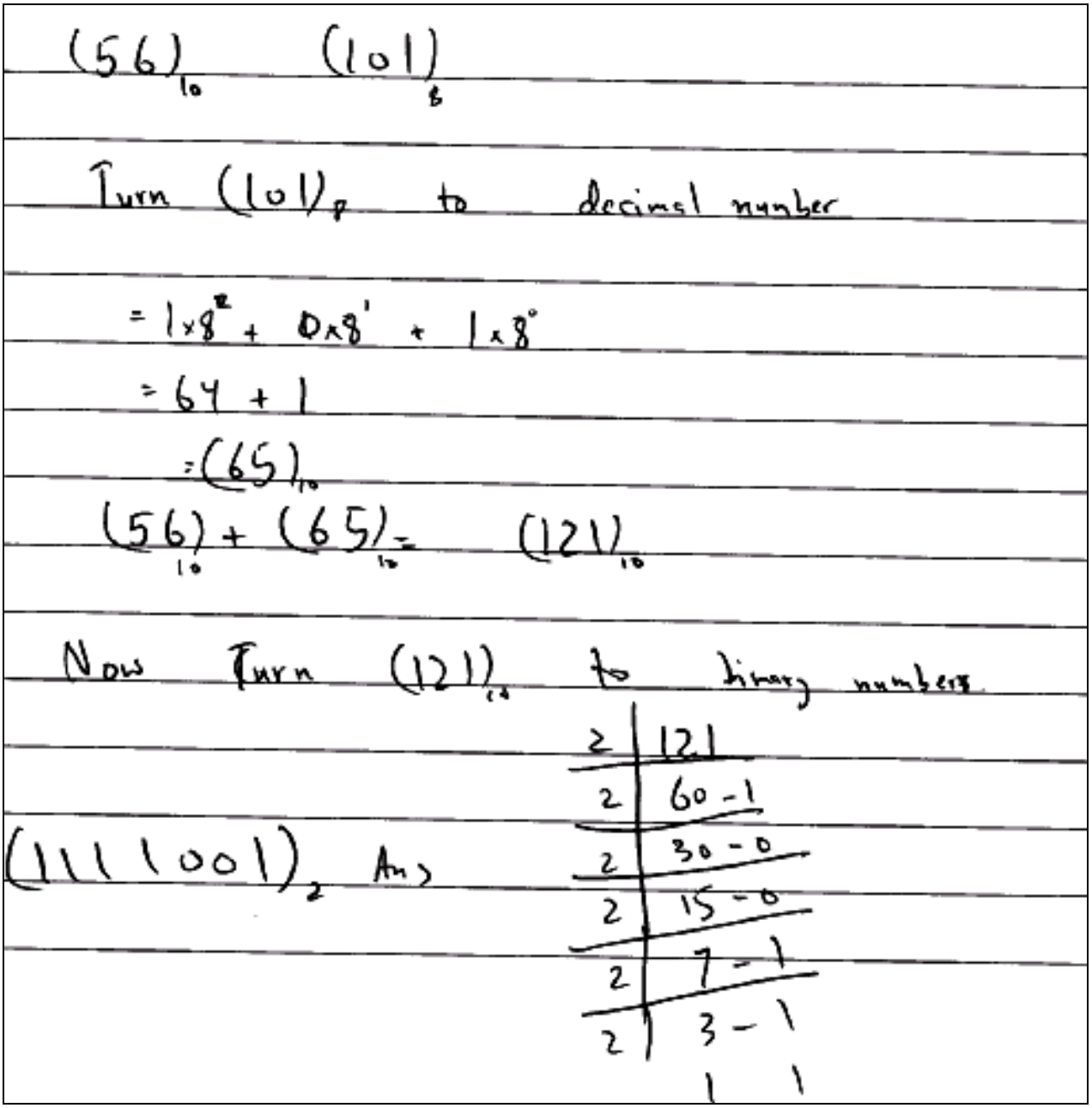


Question 3b:

Without using a calculator, express the sum of $(56)_{10}$ and $(101)_8$ in a binary number system.

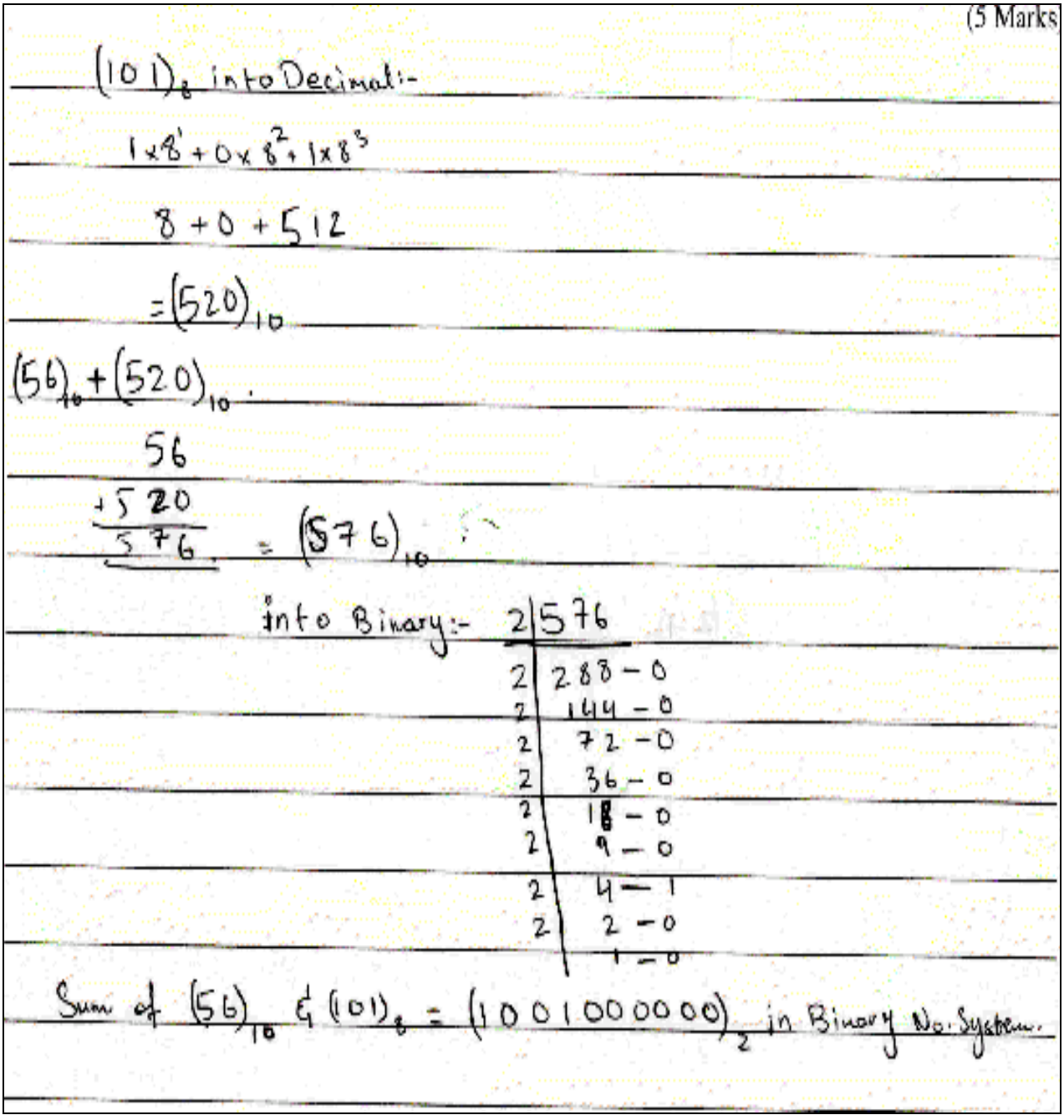
Better responses indicated that the candidates were able to understand and apply their skills in the operations of addition on different number systems. They converted $(101)_8$ to their decimal equivalents and then performed the required operation skilfully to get the answer.

Example:



Weaker responses indicated that the candidates failed to perform the number operations on the given numbers correctly. They failed to convert base 8 into decimal number system. It was also seen that fewer candidates directly converted $(101)_8$ to binary number system which was actually incorrect way. Because of this, they got stuck to proceed further which led to an incomplete and incorrect answer.

Example:



Question 4:

This question offered a choice between part **a** and part **b**. Both parts were almost equally attempted by the candidates.

Question 4a:

a. The equation of straight line is given as $\frac{x-1}{a}+\frac{y}{2}=10$.

i. Reduce this equation in slope – intercept form, i.e. $y=mx+c$.

ii. Hence, find the slope if y – intercept is 18.

Better responses indicated that in both the parts candidates had a good understanding of concepts based on linear equation. This helped candidate to convert the given equation in slope intercept form quite easily. As a result, they were able to find the slope by comparing y - intercept.

Example:

i. Reduce this equation in slope – intercept form, i.e. $y=mx+c$. (2 Marks)

$\frac{x-1}{a}+\frac{y}{2}=10$

$\frac{2(x-1)+ay}{2a}=10$

$2x-2+ay=20a$

$2x-2=20a-ay$

$2x-2=a(20-y)$

$\frac{2x-2}{a}=20-y$

$y=\frac{-2x}{a}+\frac{2+20a}{a}+20$

$y=\frac{-2x}{a}+\frac{2+20a}{a}$

$y=\frac{-2}{a}x+\frac{2+20a}{a}$

ii. Hence, find the slope if y – intercept is 18. (3 Marks)

$y=\underbrace{\frac{-2}{a}}_{\text{slope}}x+\underbrace{\frac{2+20a}{a}}_{y\text{-intercept}}$

$\frac{2+20a}{a}=18$

$2+20a=18a$

$20a-18a=-2$

$2a=-2 \quad a=-1$

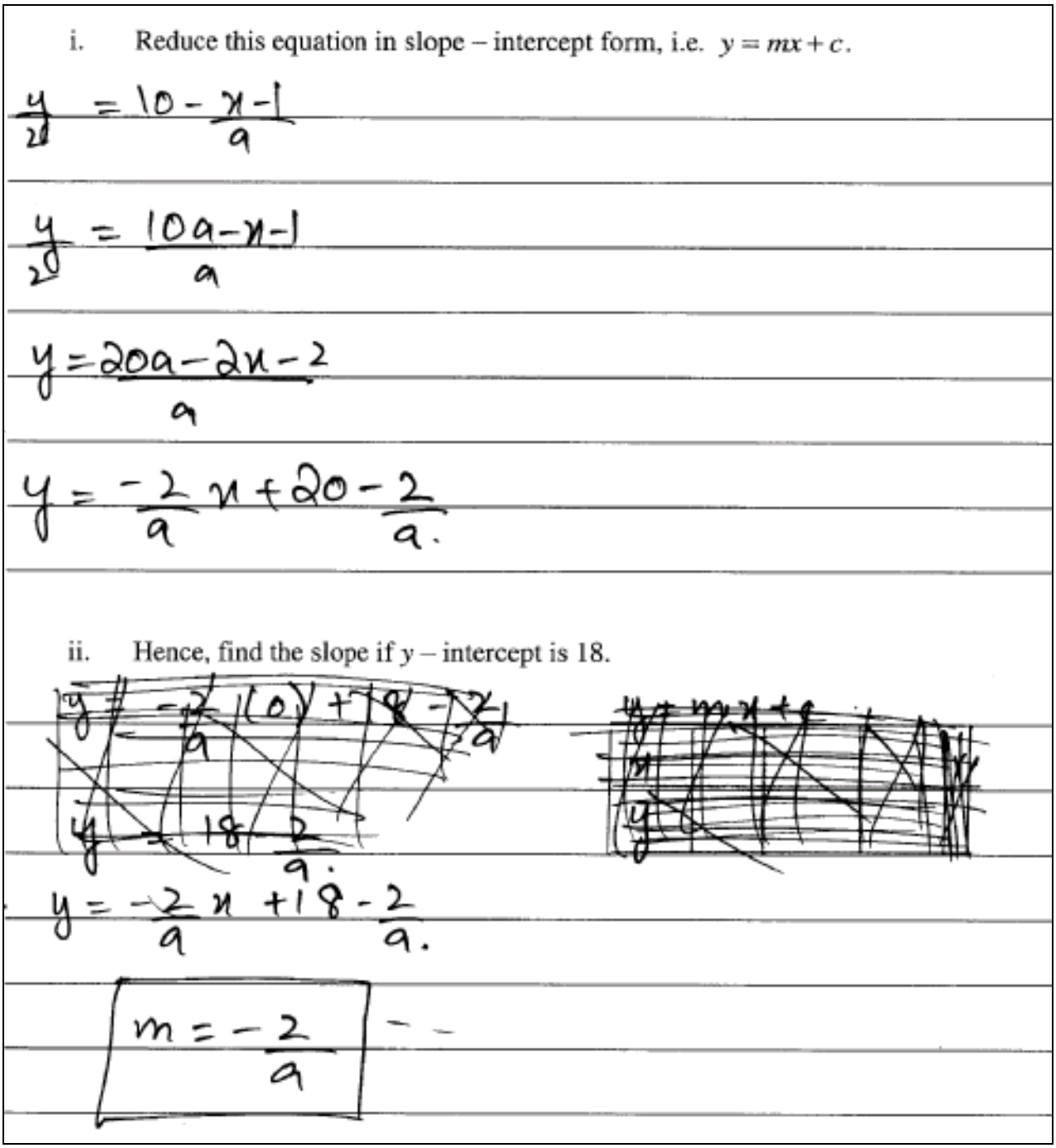
$\text{slope}=\frac{-2}{a}$

$=\frac{-2}{-1}$

$\text{slope}=2$

Weaker responses demonstrated that candidates were unable to reduce the equation in slope intercept form. They were stuck in making *y* as subject which resulted loss of marks. Because of this, they did not find the slope as well.

Example:



Question 4b:

The equation of parabola is given by $y = mx^2 + nx + r$.

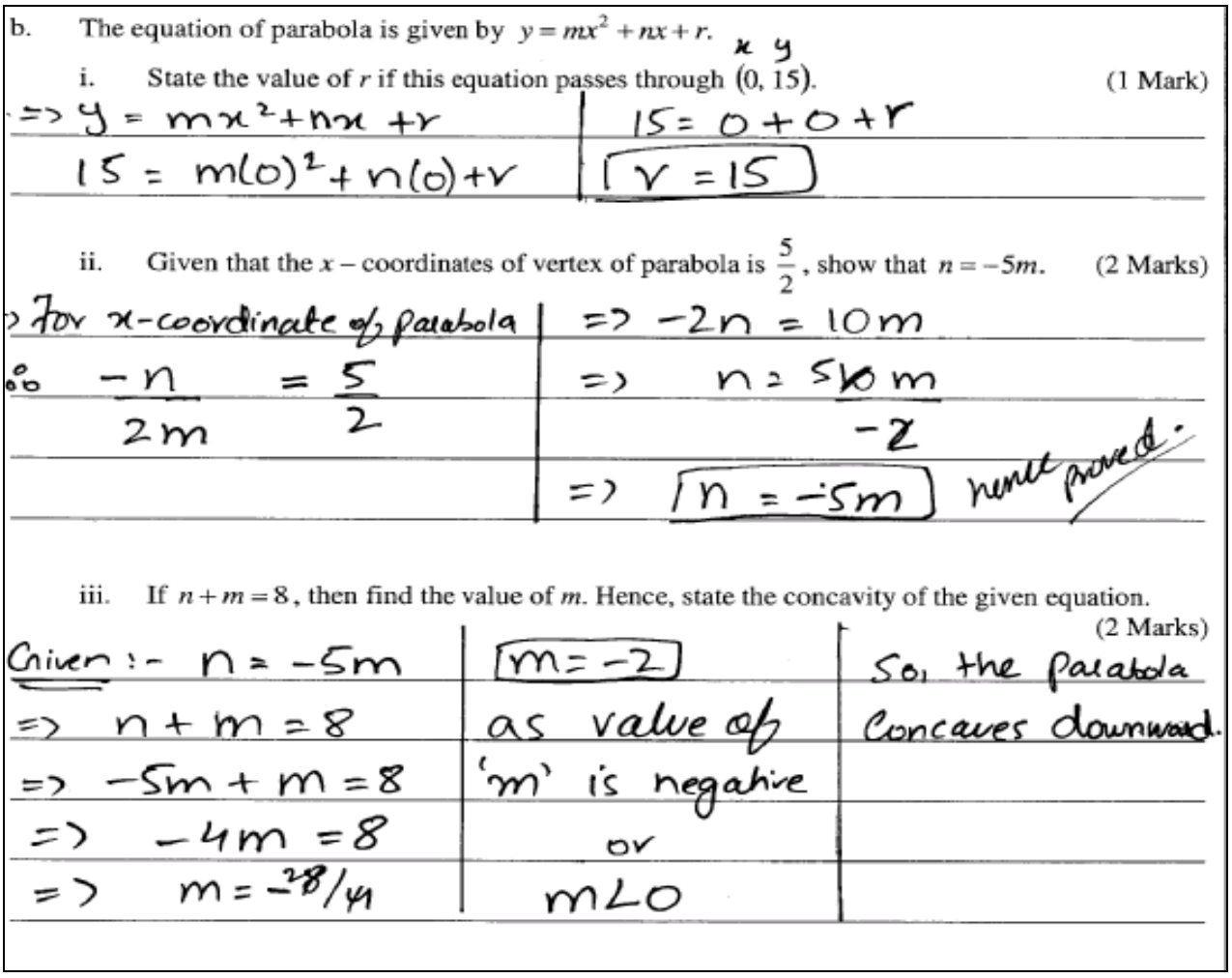
i. State the value of r if this equation passes through $(0, 15)$.

ii. Given that the x – coordinates of vertex of parabola is $\frac{5}{2}$, show that $n = -5m$.

iii. If $n + m = 8$, then find the value of m. Hence, state the concavity of the given equation.

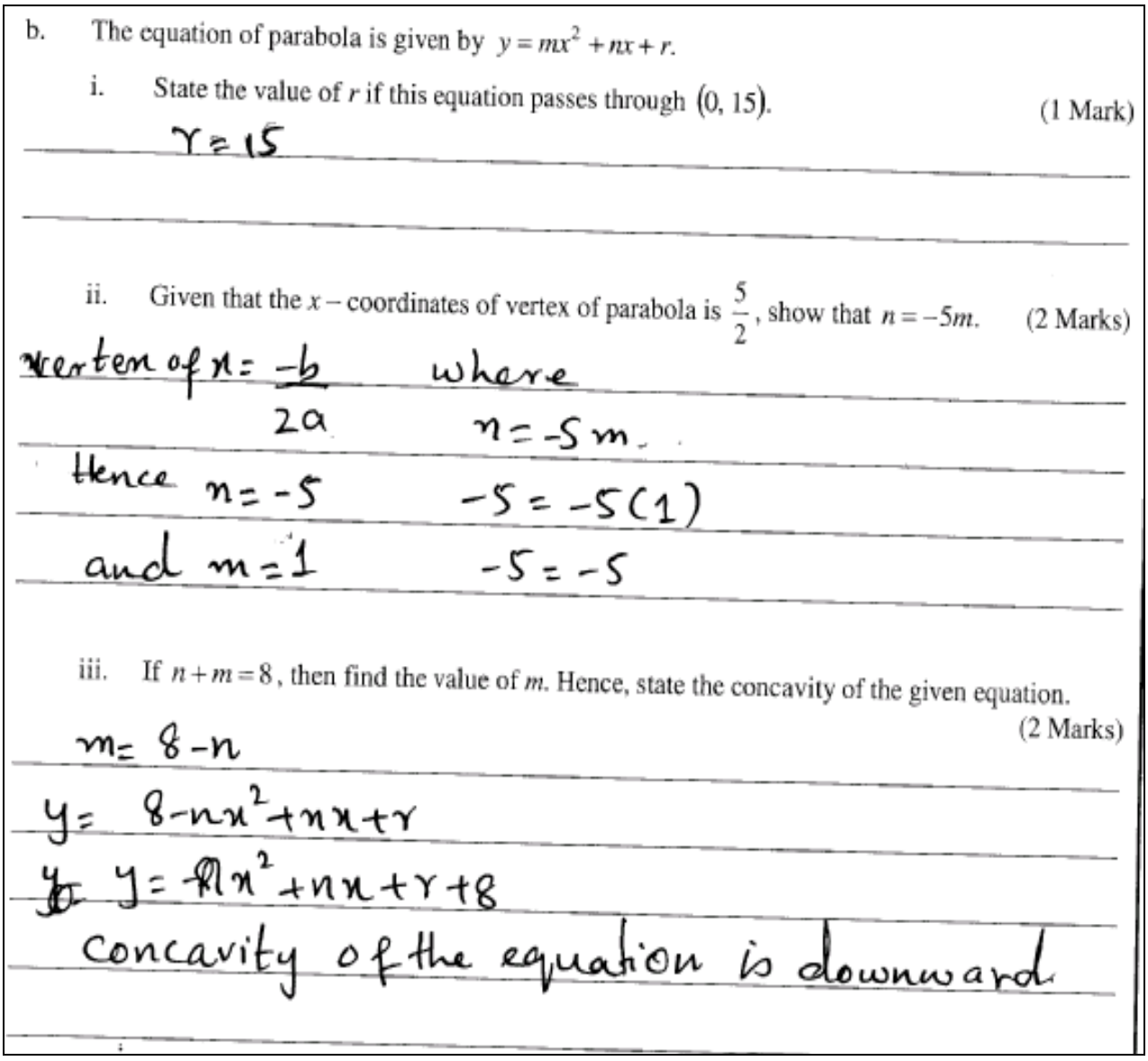
Better responses indicated that candidates, in part **i**, did well to use the given point in order to achieve the value of r. They also attained the required prove smartly in part **ii** by applying the x – coordinates of vertex, i.e. $x = -\frac{b}{2a}$. Similarly, in part **iii**, this helped candidates to get the correct value of m and so for the concavity of the given equation of parabola.

Example:



Weaker responses displayed that in both the parts candidates failed to grasp the concepts related to parabola. In part ii, some candidates were totally blank whilst in part iii, they could not use the equation properly so that they were unable to proceed further which resulted loss of marks. It was also evident that fewer candidates just substituted $\frac{5}{2}$ in the given equation of parabola and after that they were confused in proceeding further steps. As a result, they were unable to get the right way for their answers.

Example:



b. The equation of parabola is given by $y = mx^2 + nx + r$.

i. State the value of r if this equation passes through $(0, 15)$. (1 Mark)

$r = 15$

ii. Given that the $x-$coordinates of vertex of parabola is $\frac{5}{2}$, show that $n = -5m$. (2 Marks)

vertex of $x = \frac{-b}{2a}$ where $n = -5m$.

Hence $n = -5$ $\quad -5 = -5(1)$

and $m = 1$ $\quad -5 = -5$

iii. If $n + m = 8$, then find the value of m. Hence, state the concavity of the given equation. (2 Marks)

$m = 8 - n$

$y = 8 - nx^2 + nx + r$

$y = mx^2 + nx + r + 8$

concavity of the equation is downward

Question 5:

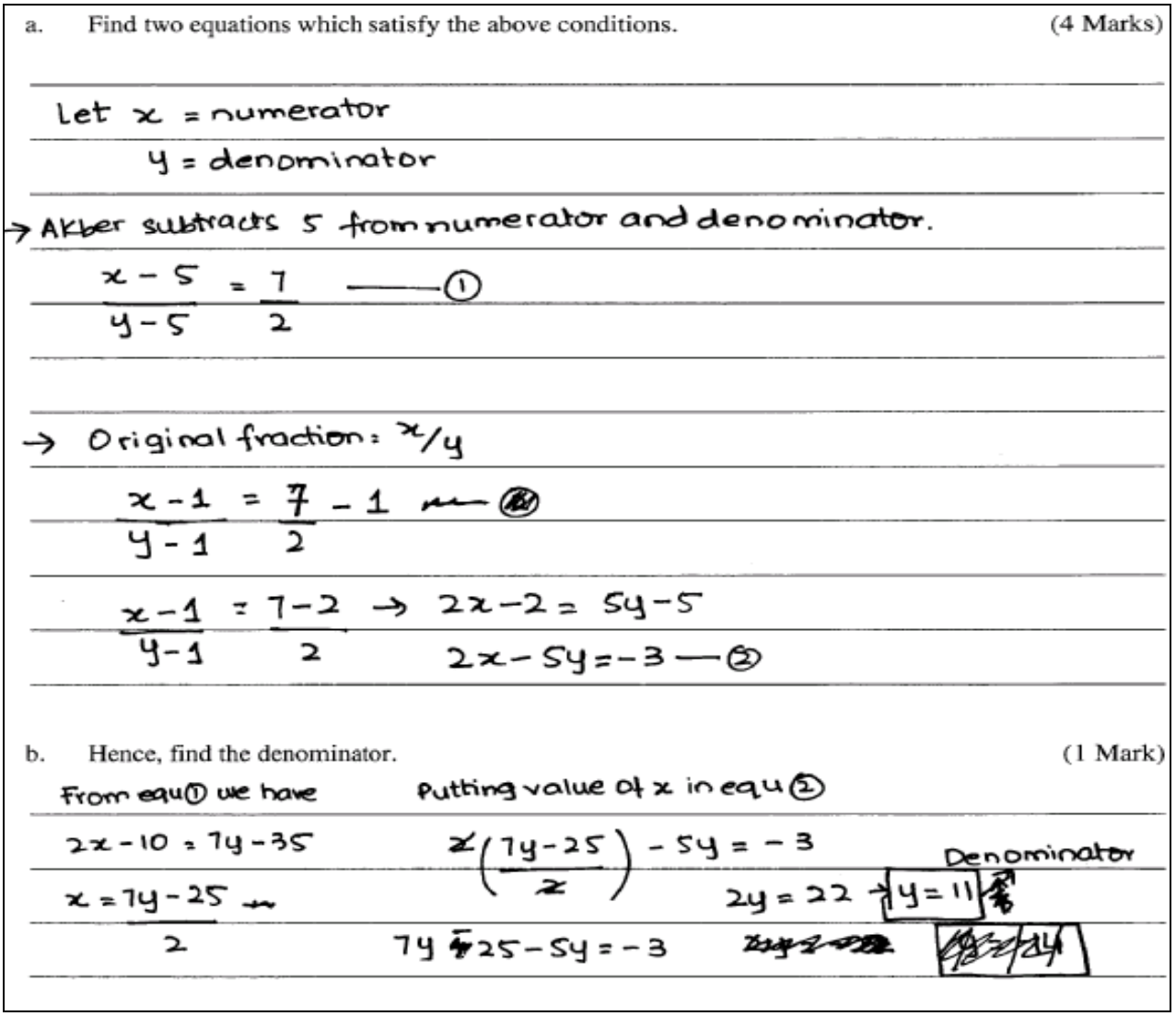
Akber subtracts 5 from the numerator and denominator of a fraction that results in $\frac{7}{2}$. This result is decreased by 1 when he adds -1 to numerator and denominator of the original fraction.

(**Note**: Let *x* be the numerator and *y* be the denominator)

a. Find two equations which satisfy the above conditions.
b. Hence, find the denominator.

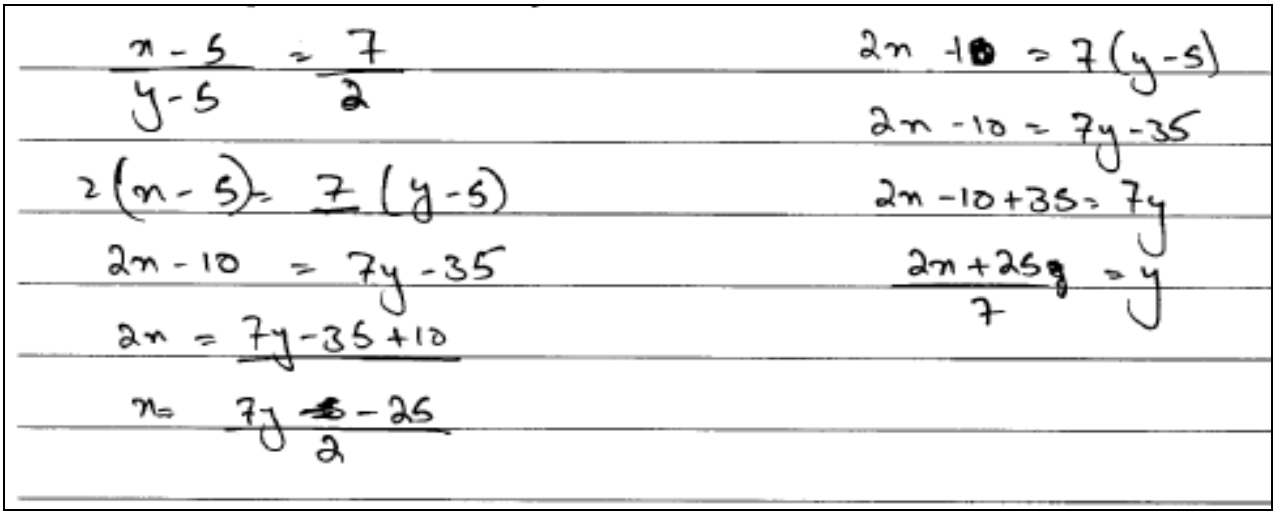
Better responses reported that candidates applied the condition very well to translate the given situation mathematically. This assisted candidates to find the denominator aptly.

Example:



Weaker responses showed that the candidates were unable to translate and used the given situation accordingly. In some cases, though they wrote one equation correctly but, could not go correctly in other equation to get the desired answer which indicated the lack of practice.

Example:



Question 6:

Let $X = [a \quad b \quad c]$, $Y = [3 \quad 4]$ and $Z = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 6 & 4 \end{bmatrix}$

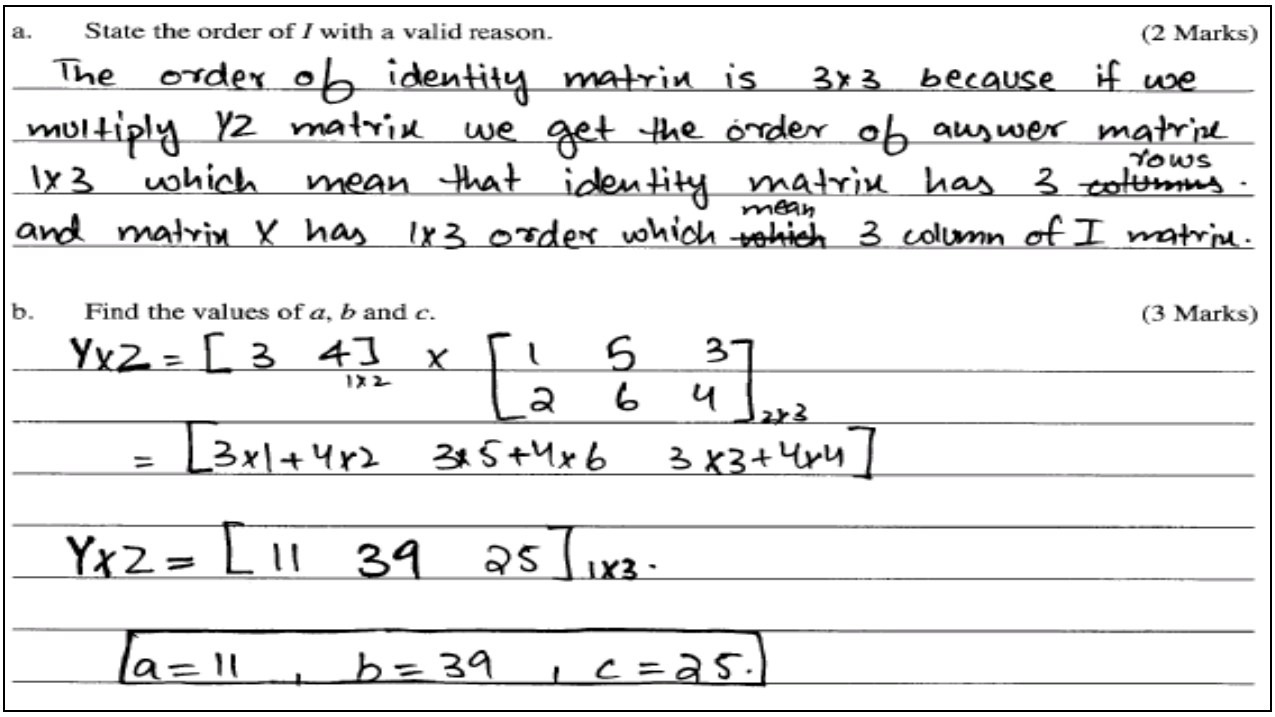
The matrices X, Y and Z are given in such a way that $YZ = XI$, where I is an identity matrix.

a. State the order of I with a valid reason.
b. Find the values of a, b and c.

Better responses demonstrated that candidates were well-versed in solving the problem based on the concept of matrices operations. In part **i**, candidates were well aware of multiplication of two matrices and because of this, they were able to write the correct reason for stating the order of identity matrix.

Similarly, in part **ii**, they performed smartly in multiplication of two matrices and finally compared the matrix $[11 \quad 39 \quad 25]$ with $[a \quad b \quad c]$. As a result, they found the required values.

Example:


a. State the order of I with a valid reason. (2 Marks)

The order of identity matrix is 3x3 because if we multiply Y2 matrix we get the order of answer matrix 1x3 which mean that identity matrix has 3 ~~columns~~ rows. and matrix X has 1x3 order which ~~which~~ mean 3 column of I matrix.

b. Find the values of a, b and c. (3 Marks)

$$Yx2 = \underset{1\times2}{[3\ \ 4]} \times \underset{2\times3}{\begin{bmatrix} 1 & 5 & 3 \\ 2 & 6 & 4 \end{bmatrix}}$$

$$= [3\times1+4\times2 \quad 3\times5+4\times6 \quad 3\times3+4\times4]$$

$$Yx2 = [11 \quad 39 \quad 25]_{1\times3}.$$

$$a = 11,\ b = 39,\ c = 25.$$

Weaker responses displayed that the candidates failed to attempt both the parts because of vague understanding and lack of practice as well. In part i, they were unable to give the right reason of stating the order of identity matrix.

Similarly, in part ii, it was noted that candidates were perplexed in multiplication of two matrices. Also, they made mistakes in comparing the corresponding entries which resulted the loss of marks.

Example:

a. State the order of I with a valid reason. (2 Marks)

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

The order of I is 2 × by 2 because indentity matrix has 1 in diagonal and all elements are zero.

b. Find the values of a, b and c. (3 Marks)

$YZ = XI$

$\begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 5 & 3 \\ 2 & 6 & 4 \end{bmatrix} = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 3(1)+4(2) & 3(5)+4(6) & 3(3)+4(4) \end{bmatrix} = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 3+8 & 15+12 & 9+16 \end{bmatrix} = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 11 & 27 & 25 \end{bmatrix} = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

The order of matrix $\begin{bmatrix} a & b & c \end{bmatrix}$ & $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is not same so multiply can not be done so

$a = 11$, $b = 27$, $c = 25$